Does the data/MC pT slope go away in Z + 1 jet?

Summer22: Z + at least 1 jet compared with Z + exactly 1 jet

Same 20 files per sample, 10 GeV pT bins, DY Z pT weights still on

2 November 2026

Slope of data/MC, in % per 100 GeV

Straight-line fit to data/MC in 10 GeV bins. The exactly-1-jet events are about 80% of the at-least-1-jet events, so the two columns are not independent.

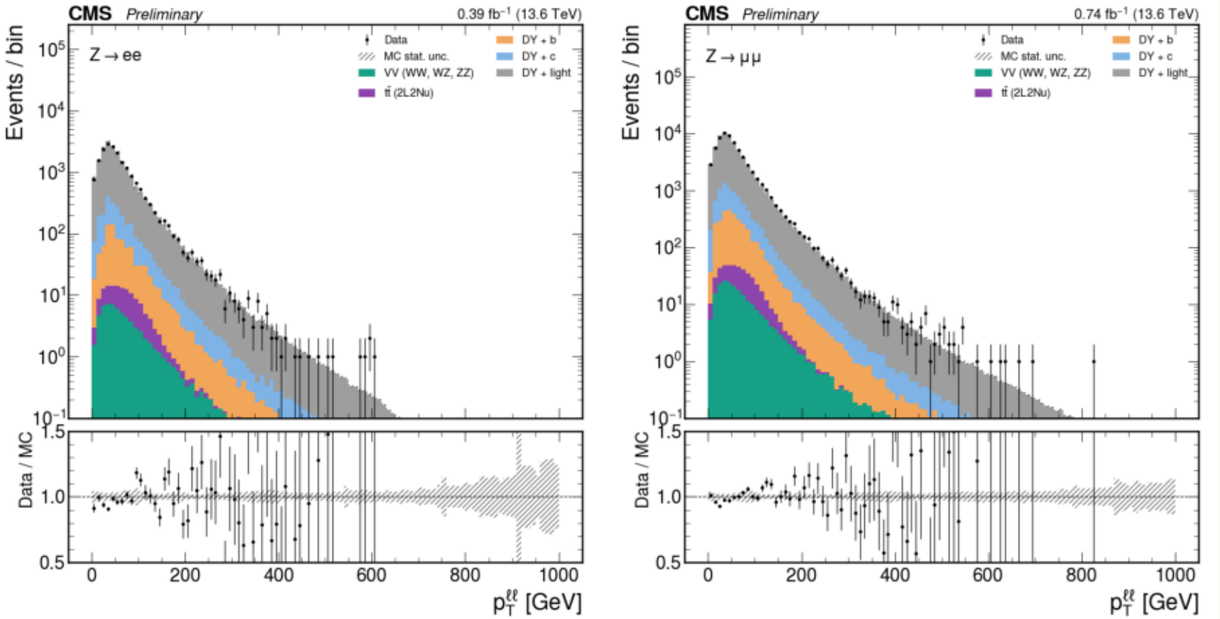
	N jet ≥ 1	N jet = 1
Z→μμ, Z pT (0–400)	+4.4 ± 1.3	+0.7 ± 1.7
Z→μμ, leading lepton (35–300)	+6.2 ± 2.2	+2.5 ± 3.0
Z→μμ, second lepton (25–200)	+20.5 ± 5.5	+1.3 ± 7.6
Z→μμ, leading jet (30–300)	+2.0 ± 1.8	−2.6 ± 2.1
Z→ee, Z pT (0–400)	+4.8 ± 2.0	+3.0 ± 2.7
Z→ee, leading lepton (35–300)	+9.7 ± 3.5	+9.1 ± 4.8
Z→ee, second lepton (25–200)	−0.9 ± 8.6	−6.3 ± 11.9
Z→ee, leading jet (30–300)	+2.3 ± 2.6	−0.4 ± 3.2

Muons: the slope is gone in Z + 1 jet. Electrons: no clear change.

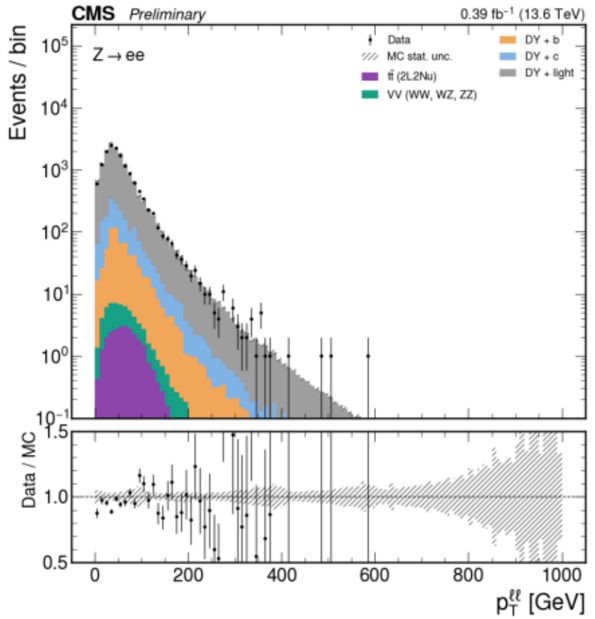
For muons the rise seems to come from events with 2 or more jets. For electrons the leading-lepton slope stays at about +9%, but the errors are large.

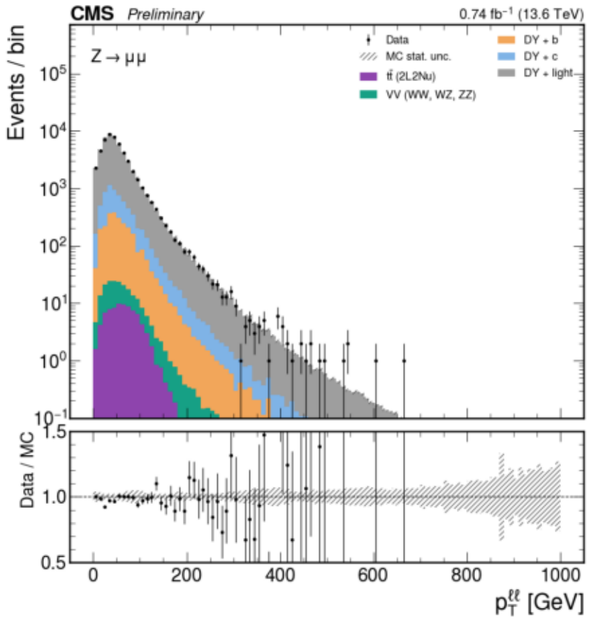
Z pT: ≥ 1 jet (left), exactly 1 jet (right)

N jet ≥ 1



N jet = 1



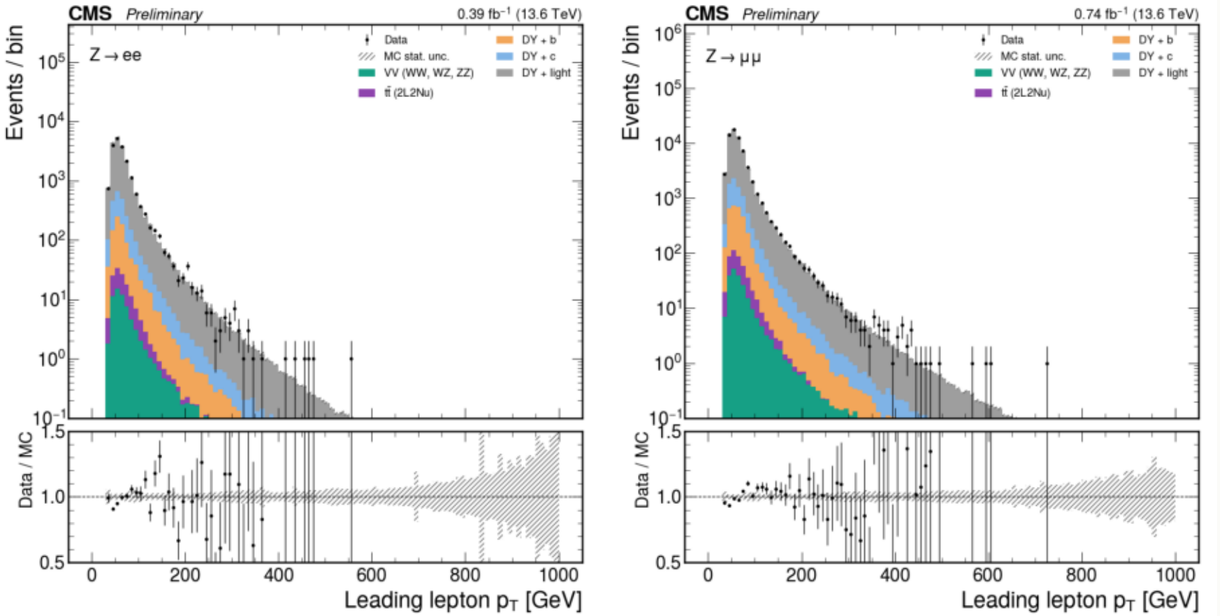


Slope: Z→μμ: +4.4 ± 1.3 → +0.7 ± 1.7. Z→ee: +4.8 ± 2.0 → +3.0 ± 2.7.

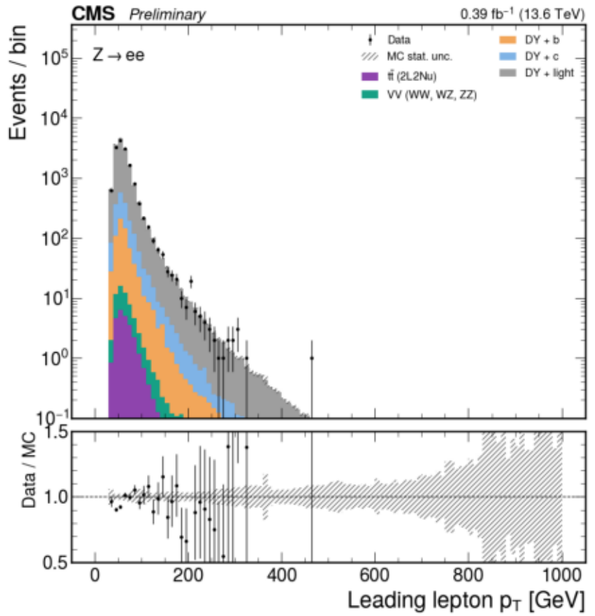
The Z→μμ rise above 100 GeV (1.04–1.06) drops to 0.99–1.03 with exactly 1 jet. Below 30 GeV data/MC is 0.95 in both selections.

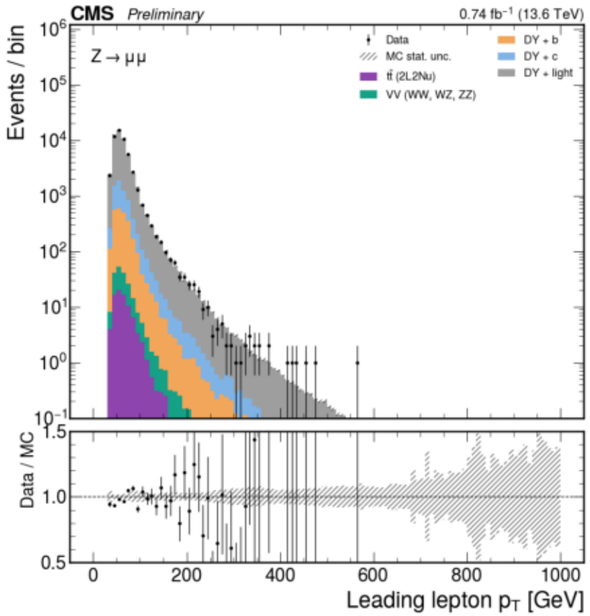
Leading-lepton pT: ≥ 1 jet (left), exactly 1 jet (right)

N jet ≥ 1



N jet = 1



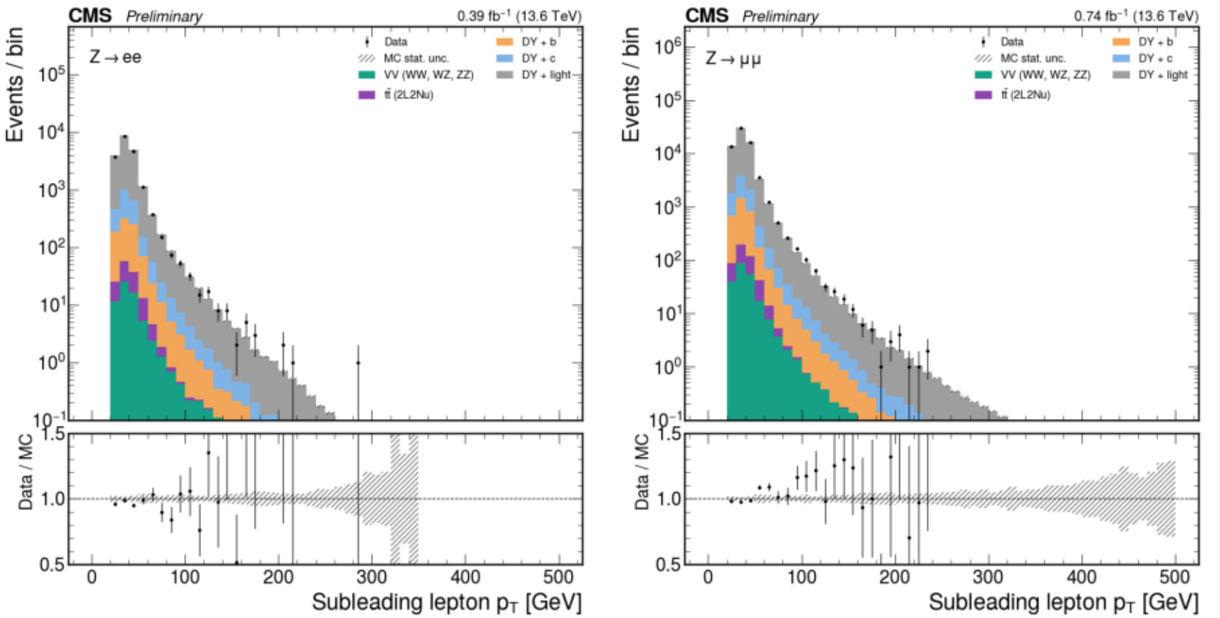


Slope: Z→μμ: +6.2 ± 2.2 → +2.5 ± 3.0. Z→ee: +9.7 ± 3.5 → +9.1 ± 4.8.

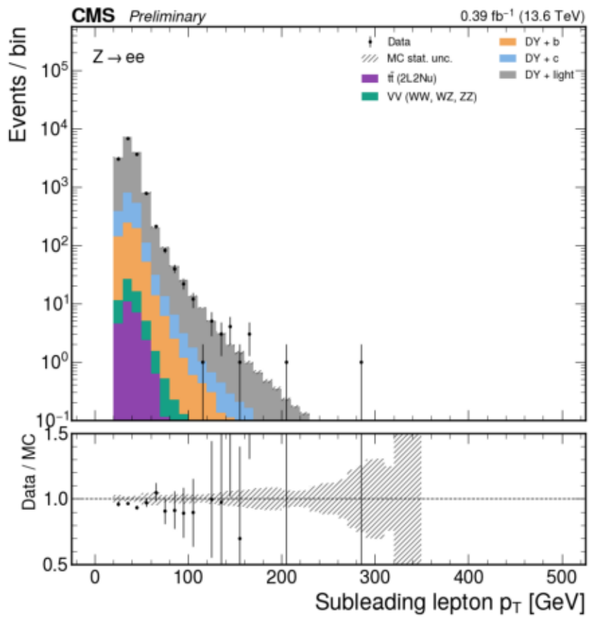
Z→μμ 100–150 GeV: 1.06 → 1.01. Z→ee does not change much.

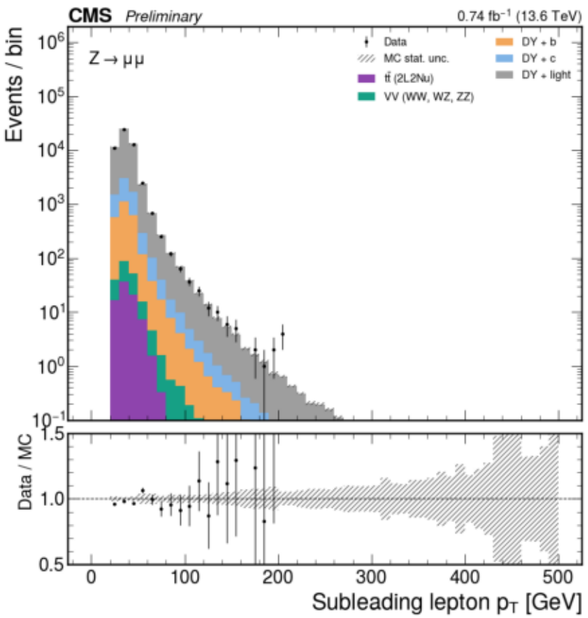
Second-lepton pT: ≥ 1 jet (left), exactly 1 jet (right)

N jet ≥ 1



N jet = 1



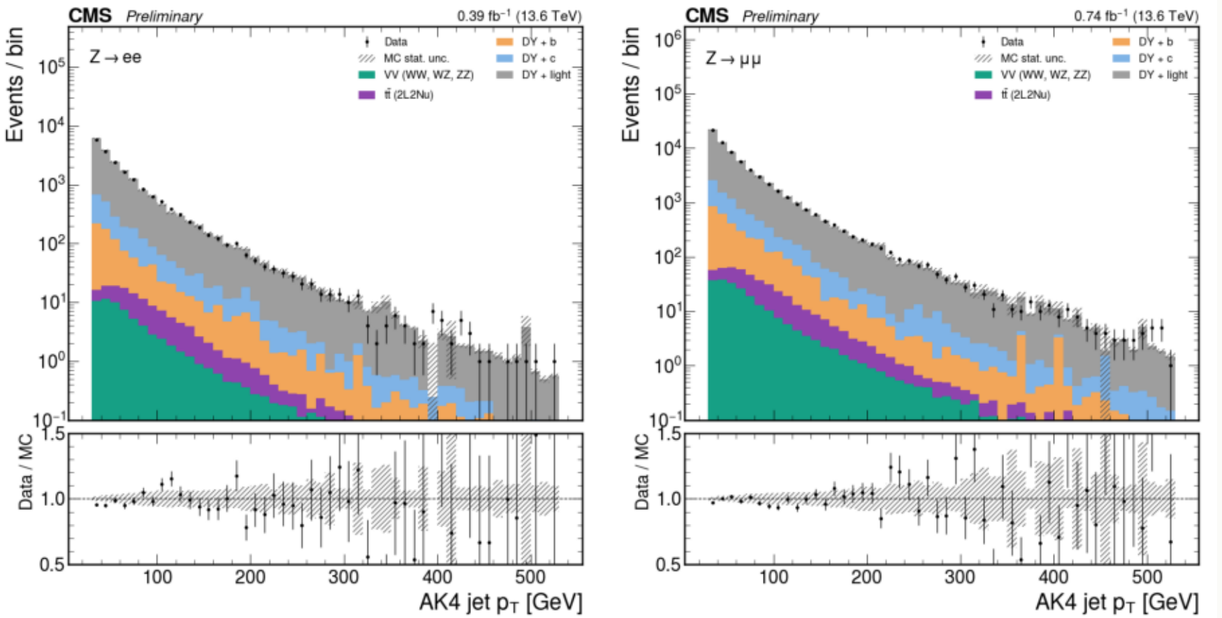


Slope: Z→μμ: +20.5 ± 5.5 → +1.3 ± 7.6. Z→ee: −0.9 ± 8.6 → −6.3 ± 11.9.

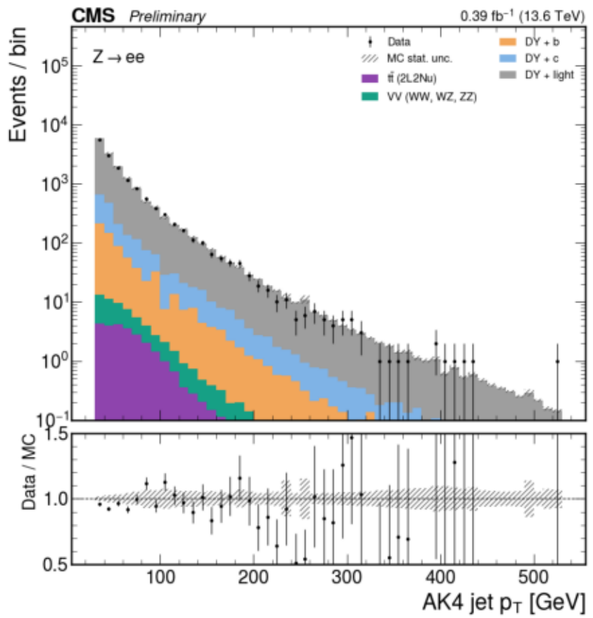
Z→μμ 50–150 GeV: 1.08–1.17 → 1.02–1.03. Electrons have too few events above 100 GeV to say anything.

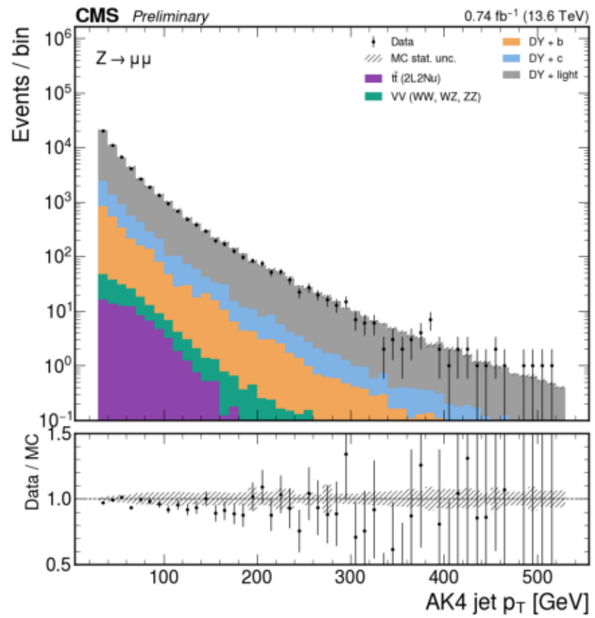
Leading-jet pT: ≥ 1 jet (left), exactly 1 jet (right)

N jet ≥ 1



N jet = 1



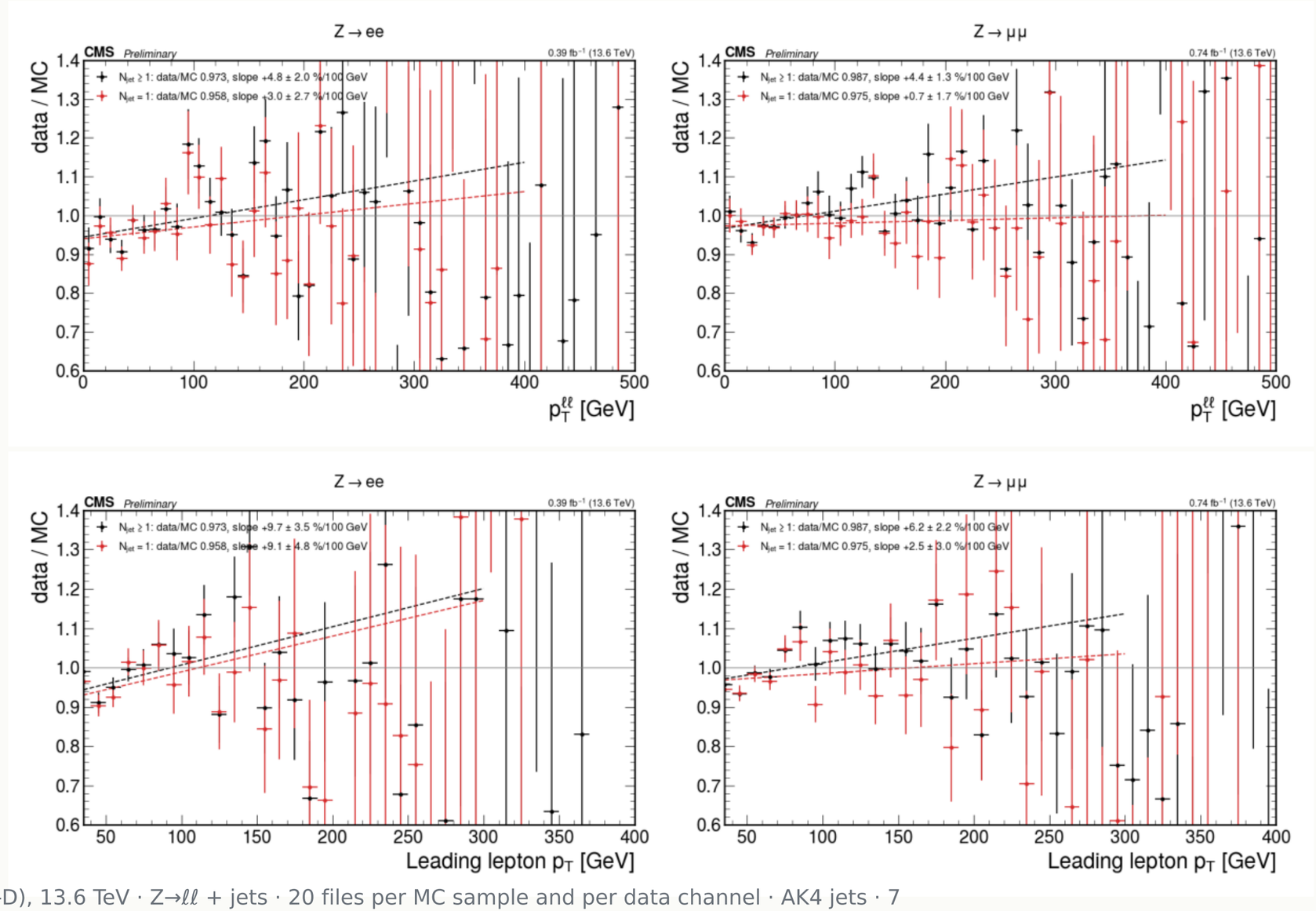


Slope: Z→μμ: +2.0 ± 1.8 → −2.6 ± 2.1. Z→ee: +2.3 ± 2.6 → −0.4 ± 3.2.

No real slope in either selection. With exactly 1 jet, MC is a bit high above 150 GeV.

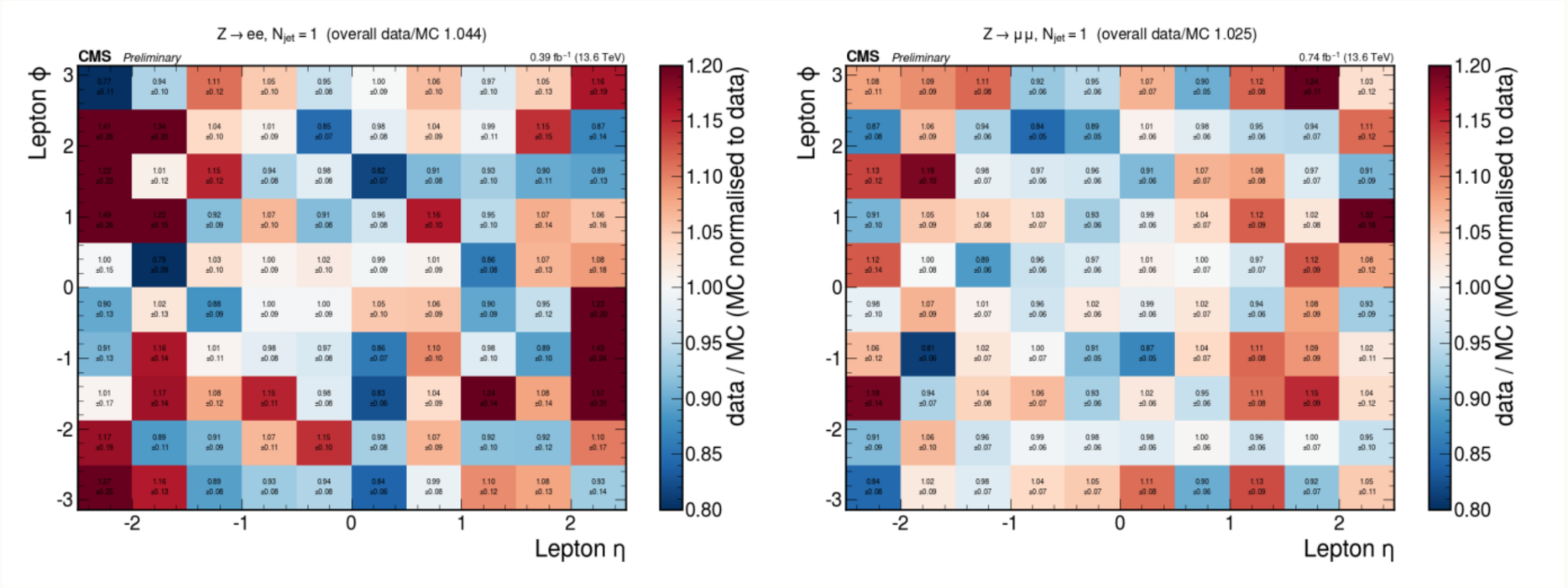
Data/MC with the fitted lines

Black: at least 1 jet. Red: exactly 1 jet.



Lepton η–φ, data/MC, Z + exactly 1 jet

MC normalised to data in each map. Each event enters twice (both leptons).



cell	N jet = 1	N jet = 2	N jet ≥ 1
Z→ee, η 0–0.5, φ 1.3–1.9	0.82 ± 0.07	0.61 ± 0.10	0.80 ± 0.06
Z→μμ, η 0.5–1.0, φ −3.1 to −2.5	0.90 ± 0.06	0.65 ± 0.08	0.85 ± 0.05

Two cells are low with 1 jet and with 2 jets.

These are separate events, so this is not one fluctuation. The rest of each map is flat (χ^2/ndf 112 and 119 for 99 with exactly 1 jet).

Summary

Muons: the pT slope goes away in Z + exactly 1 jet.

Z pT slope +4.4 ± 1.3 → +0.7 ± 1.7 %/100 GeV. The rise seems to come from events with 2 or more jets.

Electrons: no clear change.

Leading-lepton slope +9.7 ± 3.5 → +9.1 ± 4.8 %/100 GeV. We need more files to say more.

Two low η–φ cells, one per channel, low with 1 jet and with 2 jets.

Z→ee η 0–0.5, φ 1.3–1.9: 0.82 and 0.61. Z→μμ η 0.5–1.0, φ −3.1 to −2.5: 0.90 and 0.65. MC is also about 20% high for N jet ≥ 3.

Next

Same check in 2022 E–G and 2023. Rerun without the DY Z pT weights (not decided yet).